

Mathematical Foundations of Political Methodology

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- Determinant
- Eigenvalues and Eigenvectors
- Multivariate Function
- Multivariate Derivative

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Multivariable Calculus

Functions of many variables:

- 1) Policies may be multidimensional (policy provision and pork buy off)
- 2) Countries may invest in offensive and defensive resources for fighting wars
- 3) Ethnicity and resources could affect investment

Today:

- 1) Determinant
- 2) Eigenvector/Diagonalization
- 3) Multivariate functions
- 4) Partial Derivatives, Gradients, Jacobians, and Hessians

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Determinant

Suppose we have a (2×2) matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

The determinant of A for this 2×2 matrix, which is written as $|A|$ or $\det(A)$, is defined as: $a_{11}a_{22} - a_{21}a_{12}$.

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Now, suppose we have a $n \times n$ matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Define ij-entry, written a_{ij} , as the number in the i-th row and j-th column.

Define ij-minor, written $|A_{ij}|$, is the determinant that's left after deleting from $|A|$ the row and column containing a_{ij} .

Define ij-cofactor, written as A_{ij} , is given as a formula by $A_{ij} = (-1)^{i+j} |A_{ij}|$.

Now, the $\det(A)$ is defined as: $\det(A) \text{ (or } |A|) = \sum_{j=1}^n a_{ij} A_{ij} = \sum_{i=1}^n a_{ij} A_{ij}$

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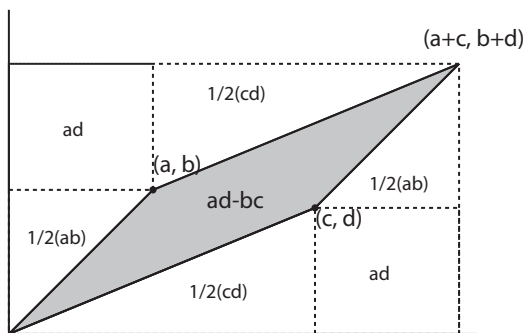
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For 2×2 matrices they have a natural interpretation:



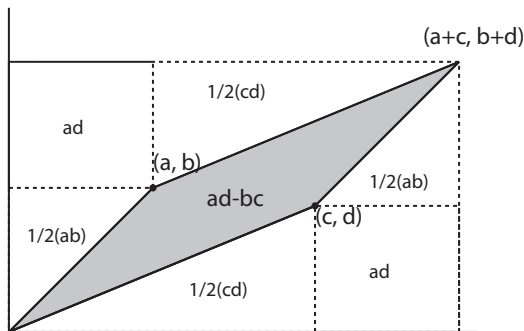
The area of the parallelogram is

$$|(a+c)(b+d) - 2ad - ab - cd| = |ab + ad + bc + cd - 2ad - ab - cd| = |ad - bc|$$

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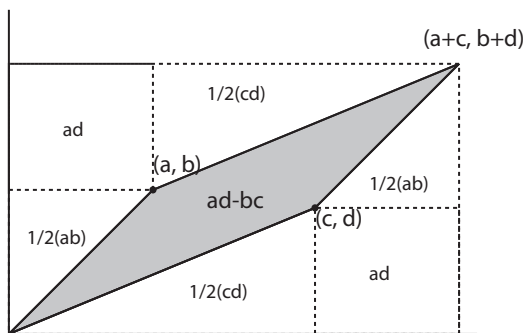
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$$A = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix}$$

Then, the determinant by expanding the first row is:

$$|A| = 1 \times \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} - 0 \times \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} + 3 \times \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$$|A| = 1 \times (-1) - 0 \times 1 + 3 \times (-3) = -10$$

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An Introduction to Eigenvectors, Values, and Diagonalization

Definition

Suppose $\mathbf{A}$ is an $N \times N$ matrix and λ is a scalar.
If

$$\mathbf{Ax} = \lambda \mathbf{x}$$

Then $\mathbf{x}$ is an eigenvector and λ is the associated eigenvalue

- $\mathbf{A}$ stretches the eigenvector $\mathbf{x}$
- $\mathbf{A}$ stretches $\mathbf{x}$ by λ
- To find eigenvectors/values: (eigen in R)
 - Find λ that solves $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$
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After we define invertibility, eigenvalues and eigenvectors, a natural question would be: what is the relationship between those concepts.

Following theorem gives us the answer.

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Theorem

Suppose $\mathbf{A}$ is an invertible $N \times N$ matrix and further suppose that $\mathbf{A}$ has N distinct eigenvalues and N linearly independent eigenvectors. Then we can write $\mathbf{A}$ as,

$$\mathbf{A} = \mathbf{W} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{pmatrix} \mathbf{W}^{-1}$$

where $\mathbf{W} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_N)$ is an $N \times N$ matrix with the N eigenvectors as column vectors.

Proof:
Note

$$\begin{aligned} \mathbf{A}\mathbf{W} &= (\lambda_1 \mathbf{w}_1 \quad \lambda_2 \mathbf{w}_2 \quad \dots \quad \lambda_N \mathbf{w}_N) \\ &= \mathbf{W}\mathbf{\Lambda} \\ \mathbf{A} &= \mathbf{W}\mathbf{\Lambda}\mathbf{W}^{-1} \end{aligned}$$

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Examples of Diagonalization

Suppose $\mathbf{A}$ is an $N \times N$ invertible matrix with eigenvalues $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_N)$ and eigenvectors $\mathbf{W}$. Calculate $\mathbf{A}\mathbf{A} = \mathbf{A}^2$

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Multivariate Functions

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$$f(x_1, x_2) = x_1^2 + x_2^2$$

Multivariate Functions

$$f(x_1, x_2) = \sin(x_1) \cos(x_2)$$

Multivariate Functions

$$f(x_1, x_2) = -(x - 5)^2 - (y - 2)^2$$

Multivariate Functions

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Multivariate Functions

$$\begin{aligned} f(\mathbf{x}) &= f(x_1, x_2, \dots, x_N) \\ &= x_1 + x_2 + \dots + x_N \\ &= \sum_{i=1}^N x_i \end{aligned}$$

Multivariate Functions

Definition

Suppose $f : R^n \rightarrow R^1$. We will call f a multivariate function. We will commonly write,

$$f(\mathbf{x}) = f(x_1, x_2, x_3, \dots, x_n)$$

- $R^n = R \underbrace{\times}_{\text{cartesian}} R \times R \times \dots R$

- The function we consider will take n inputs and output a single number (that lives in R^1 , or the real line)

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Example 1

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Evaluate at $\mathbf{x} = (x_1, x_2, x_3) = (2, 3, 2)$

$$\begin{aligned} f(2, 3, 2) &= 2 + 3 + 2 \\ &= 7 \end{aligned}$$

$$f(x_1, x_2) = x_1 + x_2 + x_1 x_2$$

Evaluate at $\mathbf{w} = (w_1, w_2) = (1, 2)$

$$\begin{aligned} f(w_1, w_2) &= w_1 + w_2 + w_1 w_2 \\ &= 1 + 2 + 1 \times 2 \\ &= 5 \end{aligned}$$

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Preferences for Multidimensional Policy

Recall that in the spatial model, we suppose policy and political actors are located in a space.

Suppose that policy is N dimensional—or $\mathbf{x} \in \mathbb{R}^N$.

Suppose that legislator i 's utility is a $U : \mathbb{R}^N \rightarrow \mathbb{R}^1$ and is given by,

$$\begin{aligned}U(\mathbf{x}) &= U(x_1, x_2, \dots, x_N) \\&= -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2 - \dots - (x_N - \mu_N)^2 \\&= -\sum_{j=1}^N (x_j - \mu_j)^2\end{aligned}$$

Suppose $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_N) = (0, 0, \dots, 0)$. Evaluate legislator's utility for a policy proposal of $\mathbf{m} = (1, 1, \dots, 1)$.

$$\begin{aligned}U(\mathbf{m}) &= U(1, 1, \dots, 1) \\&= -(1 - 0)^2 - (1 - 0)^2 - \dots - (1 - 0)^2 \\&= -\sum_{j=1}^N 1 = -N\end{aligned}$$

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- Determinant
- Eigenvalues and Eigenvectors
- Multivariate Function
- Multivariate Derivative

Multivariate Derivative

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, where $X \subset \mathbb{R}^n$. $f(\mathbf{x}) = f(x_1, x_2, \dots, x_N)$. If the limit,

$$\begin{aligned}\frac{\partial}{\partial x_i} f(\mathbf{x}_0) &= \frac{\partial}{\partial x_i} f(x_{01}, x_{02}, \dots, x_{0i}, x_{0i+1}, \dots, x_{0N}) \\ &= \lim_{h \rightarrow 0} \frac{f(x_{01}, x_{02}, \dots, x_{0i} + h, \dots, x_{0N}) - f(x_{01}, x_{02}, \dots, x_{0i}, \dots, x_{0N})}{h}\end{aligned}$$

exists then we call this the partial derivative of f with respect to x_i at the value $\mathbf{x}_0 = (x_{01}, x_{02}, \dots, x_{0N})$.

Rules for Taking Partial Derivatives

Partial Derivative: $\frac{\partial f(\mathbf{x})}{\partial x_i}$

- Treat each instance of x_i as a variable that we would differentiate before
- Treat each instance of $\mathbf{x}_{-i} = (x_1, x_2, x_3, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ as a constant

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Example Partial Derivatives

$$\begin{aligned}f(\mathbf{x}) &= f(x_1, x_2) \\ &= x_1 + x_2\end{aligned}$$

Partial derivative, with respect to x_1 at (x_{01}, x_{02})

$$\begin{aligned}\frac{\partial f(x_1, x_2)}{\partial x_1} \Big|_{(x_{01}, x_{02})} &= 1 + 0 \Big|_{x_{01}, x_{02}} \\ &= 1\end{aligned}$$

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$$\begin{aligned}f(\mathbf{x}) &= f(x_1, x_2, x_3) \\ &= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2\end{aligned}$$

What is the partial derivative with respect to x_1 ? x_2 ? x_3 ? Evaluated at $\mathbf{x}_0 = (x_{01}, x_{02}, x_{03})$.

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$$\begin{aligned}\frac{\partial f(\mathbf{x})}{\partial x_1} \Big|_{\mathbf{x}_0} &= 2x_1 \log(x_1) + x_1^2 \frac{1}{x_1} + x_2 x_3 \Big|_{\mathbf{x}_0} \\ &= 2x_{01} \log(x_{01}) + x_{01} + x_{02} x_{03}\end{aligned}$$

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Rate of Change, Linear Model

Suppose we regress Approval_i rate for Obama in month i on Employ_i and Gas_i . We obtain the following model:

$$\text{Approval}_i = 0.8 - 0.5\text{Employ}_i - 0.25\text{Gas}_i$$

We are modeling $\text{Approval}_i = f(\text{Employ}_i, \text{Gas}_i)$. What is partial derivative with respect to employment?

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Gradient

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$ with $X \subset \mathbb{R}^n$ is a differentiable function. Define the gradient vector of f at $\mathbf{x}_0$, $\nabla f(\mathbf{x}_0)$ as,

$$\nabla f(\mathbf{x}_0) = \left(\frac{\partial f(\mathbf{x}_0)}{\partial x_1}, \frac{\partial f(\mathbf{x}_0)}{\partial x_2}, \frac{\partial f(\mathbf{x}_0)}{\partial x_3}, \dots, \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \right)$$

- The gradient points in the direction that the function is increasing in the fastest direction
- We'll use this to do optimization (both analytic and computational)

Example Gradient Calculation

Suppose

$$\begin{aligned}f(\mathbf{x}) &= f(x_1, x_2, \dots, x_n) \\&= x_1^2 + x_2^2 + \dots + x_n^2 \\&= \sum_{i=1}^n x_i^2\end{aligned}$$

Then $\nabla f(\mathbf{x}^*)$ is

$$\nabla f(\mathbf{x}^*) = (2x_1^*, 2x_2^*, \dots, 2x_n^*)$$

So if $\mathbf{x}^* = (3, 3, \dots, 3)$ then

$$\begin{aligned}\nabla f(\mathbf{x}^*) &= (2 * 3, 2 * 3, \dots, 2 * 3) \\&= (6, 6, \dots, 6)\end{aligned}$$

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Second Partial Derivative

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Suppose $f : X \rightarrow \mathbb{R}$ where $X \subset \mathbb{R}^n$ and suppose that $\frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_i}$ exists. Then we define,

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} \equiv \frac{\partial}{\partial x_j} \left(\frac{\partial f(\mathbf{x})}{\partial x_i} \right)$$

- Second derivative could be with respect to x_i or with some other variable x_j
- Nagging question: does order matter?

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Second Partial Derivative: Order Doesn't Matter

Theorem

Young's Theorem Let $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^n$ be a twice differentiable function on all of X . Then for any i, j , at all $\mathbf{x}^* \in X$,

$$\frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x}^*) = \frac{\partial^2}{\partial x_j \partial x_i} f(\mathbf{x}^*)$$

Second Order Partial Derivates

$$f(\mathbf{x}) = x_1^2 x_2^2$$

Then,

$$\frac{\partial^2}{\partial x_1 \partial x_1} f(\mathbf{x}) = 2x_2^2$$

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$$\frac{\partial^2}{\partial x_1 \partial x_1} f(\mathbf{x}) = 2x_2^2$$

$$\frac{\partial^2}{\partial x_1 \partial x_2} f(\mathbf{x}) = 4x_1 x_2$$

$$\frac{\partial^2}{\partial x_2 \partial x_2} f(\mathbf{x}) = 2x_1^2$$

Hessians

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$, with f a twice differentiable function. We will define the Hessian matrix as the matrix of second derivatives at $\mathbf{x}^* \in X$,

$$\mathbf{H}(f)(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

- Hessians are symmetric
- They describe curvature of a function (think, how bended)
- Will be the basis for second derivative test for multivariate optimization

An Example

Suppose $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, with

$$f(x_1, x_2, x_3) = x_1^2 x_2^2 x_3^2$$

$$\nabla f(\mathbf{x}) = (2x_1 x_2^2 x_3^2, 2x_1^2 x_2 x_3^2, 2x_1^2 x_2^2 x_3)$$

$$H(f)(\mathbf{x}) = \begin{pmatrix} 2x_2^2 x_3^2 & 4x_1 x_2 x_3^2 & 4x_1 x_2^2 x_3 \\ 4x_1 x_2 x_3^2 & 2x_1^2 x_3^2 & 4x_1^2 x_2 x_3 \\ 4x_1 x_2^2 x_3 & 4x_1^2 x_2 x_3 & 2x_1^2 x_2^2 \end{pmatrix}$$

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Functions with Multidimensional Codomains

Definition

Suppose $f : R^m \rightarrow R^n$. We will call f a multivariate function. We will commonly write,

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{pmatrix}$$

Example Functions

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}^2$,

$$f(t) = (t^2, \sqrt{t})$$

Suppose we have some policy $\mathbf{x} \in \mathbb{R}^M$. Suppose we have N legislators where legislator i has utility

$$U_i(\mathbf{x}) = \sum_{j=1}^M -(x_j - \mu_{ij})^2$$

We can describe the utility of all legislators to the proposal as

$$f(\mathbf{x}) = \begin{pmatrix} \sum_{j=1}^M -(x_j - \mu_{1j})^2 \\ \sum_{j=1}^M -(x_j - \mu_{2j})^2 \\ \vdots \\ \sum_{j=1}^M -(x_j - \mu_{Nj})^2 \end{pmatrix}$$

Example Functions

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as

$$f(r, \theta) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$$

Suppose we have some policy $\mathbf{x} \in \mathbb{R}^M$. Suppose we have N legislators where legislator i has utility

$$U_i(\mathbf{x}) = \sum_{j=1}^M -(x_j - \mu_{ij})^2$$

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Jacobian

Definition

Suppose $f : X \rightarrow \mathbb{R}^n$, where $X \subset \mathbb{R}^m$, with f a differentiable function. Define the Jacobian of f at $\mathbf{x}$ as

$$\mathbf{J}(f)(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_m} \end{pmatrix}$$

Remark: The Jacobian matrix is important because if the function f is differentiable at a point $\mathbf{x}$, then the Jacobian matrix defines a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$, which is the best (pointwise) linear approximation of the function f near the point $\mathbf{x}$. This linear map is thus the generalization of the usual notion of derivative, and is called the derivative or the differential of f at $\mathbf{x}$.

Example of Jacobian

$$f(r, \theta) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$$

$$\mathbf{J}(f)(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$